MINGI KANG

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EDUCATION

Bowdoin College, ME *August 2022 - May 2026*
Bachelors of Arts in Computer Science, Minor in Mathematics
Overall GPA: 3.71 / 4.00

Aquincum Institute of Technology, Budapest, Hungary *August - December 2024*
Study Abroad Semester in Computer Science

PUBLICATIONS, PRESENTATIONS, MEDIA

Publications

[1] FCT: Fully Convolutional Transformers with Linear Attention 2026
Mingi Kang, Jackson Codd, Jeová Farias Sales Rocha Neto *In Preparation*

[2] IGLU: The Integrated Gaussian Linear Unit Activation Function 2026
Mingi Kang, Zai Yang, Jeová Farias Sales Rocha Neto *Under Review*
arXiv preprint arXiv:2603.06861

[3] Attention Via Convolutional Nearest Neighbors 2025
Mingi Kang, Jeová Farias Sales Rocha Neto *Preprint*
arXiv preprint arXiv:2511.14137

[4] Parallel qMRI Reconstruction from 4x Accelerated Acquisitions 2025
Mingi Kang *Technical Report*
arXiv preprint arXiv:2511.18232

[5] Structures and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing 2024
Conference Publication
Abhilasha A. Kumar, Mingi Kang, William G. Kronenberger, Michael N. Jones, David B. Pisoni
CogSci 2024. Proceedings of the 46th Annual Meeting of the Cognitive Science Society (Vol. 46).
DOI: https://escholarship.org/uc/item/7vn9q9hh

Presentations

[1] Parallel qMRI Reconstruction from 4x Accelerated Acquisitions July 2025
McKelvey School of Engineering Summer Symposium, St. Louis, MO *Oral*

[2] Convolutional Nearest Neighbors: Reinterpreting Convolution Through K-Nearest Neighbor Selection October 2025
IEEE MIT Undergraduate Research Technology Conference (MIT URTC) 2025, Cambridge, MA *Poster*

[3] Parallel qMRI Reconstruction from 4x Accelerated Acquisitions July 2025
McKelvey School of Engineering Poster Palooza, St. Louis *Poster*

[4] Structures and process-level lexical interactions in memory search: A case study of individuals with cochlear implants and normal hearing July 2024
Poster
Annual Conference of the Cognitive Science Society (CogSci) 2024, Rotterdam, Netherlands

Media

[1] Mingi Kang '26: Advancing Computers' Ability to See and Understand Our World *School Article*
Bowdoin Article Link *November 2025*

RESEARCH EXPERIENCE

Senior Honors Thesis in Computer Science August 2025 - Present
Honors Candidate (Advisor: Prof Jeova Farias) *Brunswick, ME*

- Developed **ConvNN-Triton**, a custom Triton kernel for the ConvNN neighbor selection and aggregation mechanism, achieving 1.16× faster forward pass, 1.36× faster backward pass, 2.32× lower forward peak memory, and 1.38× lower backward peak memory over baseline PyTorch implementation.
- Architected a hybrid branching layer combining spatial (Conv2d) and feature-based (ConvNN) aggregation, achieving up to 0.9% accuracy improvement on ImageNet-1K with ResNet-50 while reducing GFLOPS by 32% and parameter count by 3% compared to standard convolution.
- Extended ConvNN to an attention variant (ConvNN-Attention) that improves over ViT-Base self-attention by 0.7% on ImageNet-1K with 2% fewer GFLOPS and only 28K additional parameters.
- Designed multi-GPU training on Indiana Jetstream 4×H100 GPUs, implementing Distributed Data Parallel (DDP) and mixed-precision training for large-scale ImageNet-1K experiments across ViT and ResNet architectures.

Independent Deep Learning Research Group August 2025 - Present
Undergraduate Research Assistant (Advisor: Prof Jeova Farias) *Brunswick, ME*

- Developed **IGLU** (Integrated Gaussian Linear Unit), a parametric activation function derived as a scale mixture of GELU gates under a half-normal mixing distribution, generalizing both ReLU and GELU with a learnable parameter σ for fine-grained control over activation shape.
- Implemented a modular GPT-2 architecture with plug-and-play activation swapping, training on WikiText-103 to benchmark perplexity: IGLU achieves a 0.11 reduction in test perplexity compared to GELU.
- Demonstrated 1-2% accuracy improvements on imbalanced CIFAR-100 classification with ResNet-20 over ReLU and self-gating baselines (GELU, SiLU).

Computational Imaging Group, Washington University in St. Louis May - August 2025
McKelvey Summer Engineering Fellow (Advisor: Prof Ulugbek S. Kamilov) *St. Louis, MO*

- Designed a self-supervised training pipeline using 2× accelerated scans with k-space masking to simulate 4× undersampling, enabling model training without full-sampled ground truth – validated against real 4× accelerated acquisitions at test time.
- Redesigned a deep unfolding U-Net architecture for parallel qMRI reconstruction, achieving a 4× parameter reduction while maintaining reconstruction quality (37 dB PSNR, 0.923 SSIM) on 4× accelerated scans.
- Developed ACS region-specific and coil-instance normalization techniques for undersampled k-space preprocessing, validated on a patient dataset from WashU Medical Center.

Independent Deep Learning Research Group January - May 2025
Undergraduate Research Assistant (Advisor: Prof Jeova Farias) *Brunswick, ME*

- Developed **Convolutional Nearest Neighbor Attention (ConvNN-Attention)**, an efficient attention mechanism using hard k-NN and convolutional aggregation, reducing computational cost by 18% (GFLOPS) with accuracy improvement (2.4%) on CIFAR-10/100.
- Implemented modular PyTorch layers compatible with standard Transformer and Vision Transformer architectures, enabling drop-in replacement for self-attention.

Christenfeld Summer Research Fellowship January - August 2024
Research Fellow (Advisor: Prof Jeova Farias) *Brunswick, ME*

- Developed **Convolutional Nearest Neighbors (ConvNN)** algorithm integrating norm-based k-NN pixel selection into standard convolutional operations for spatial feature learning.
- Built modular PyTorch implementation (1D/2D) with configurable sampling strategies (random, spatial), pixel-shuffling, and positional encoding.

Lexicon Lab, Bowdoin College December 2022 - May 2024
Undergraduate Research Assistant (Advisor: Prof Abhilasha Kumar) *Brunswick, ME*

- Investigated lexical retrieval processes in prelingually deaf cochlear implant users through computational cognitive modeling, contributing to published CogSci 2024 conference proceedings paper (second author).
- Extended Python package, *Forager* with joint semantic word embeddings (word2vec, speech2vec) for quantitative analysis, revealing differential reliance on speech-derived representations.

TEACHING/MENTORING EXPERIENCE

Bowdoin College Baldwin Center for Learning and Teaching August 2025 - Present
Quantitative Tutor *Brunswick, ME*

- Provide one-on-one tutoring in computer science (algorithms, data structures, AI), mathematics (linear algebra, probability, statistics), and economics (microeconomics, macroeconomics).
- Support students in debugging code, understanding theoretical concepts, and developing problem solving strategies for quantitative coursework.

Bowdoin College Mathematics Department August 2025 - Present
Learning Assistant (Teaching Assistant) for Statistics and Data Science *Brunswick, ME*

- Conduct weekly learning assistant hours supporting 20+ students with R programming, statistical analysis, hypothesis testing, data visualization, and data cleaning.
- Guide students through applied projects involving real-world datasets, emphasizing statistical thinking and reproducible analysis.

Bowdoin College Computer Science Department January 2025 - Present
Learning Assistant (Teaching Assistant) for Intro to Computer Science *Brunswick, ME*

- Lead weekly learning assistant hours for 40+ students covering Python fundamentals and object-oriented programming.
- Mentor students on debugging techniques, code organization, and developer best practices.

Bowdoin College Career Exploration and Development Group January 2025, January 2026
Sophomore Bootcamp Leader & Pod Leader *Brunswick, ME*

- Design and deliver workshops on career development for 400+ sophomores, covering resume, writing, behavioral interviews, internship navigation, personal finance, academic research, graduate school, and fellowships.
- Train and manage a team of 7 senior mentors, overseeing their preparation to deliver specialized workshops and career curricula to the sophomore class.
- Mentored 5 student teams in inaugural Sophomore Bootcamp Hackathon, providing technical guidance on full-stack development, API integration, and project management.

RELEVANT COURSEWORK

Computer Science: Deep Learning for Computer Vision, Operating Systems, Artificial Intelligence, Computer Systems, Algorithms, Data Structures, Data Science, Cryptography, Computational Game Theory

Mathematics: Linear Algebra, Multivariable Calculus, Probability, Statistics, Mathematical Reasoning, Advanced Probability and Statistics

TECHNICAL SKILLS

Programming Languages	Python, R, Java, C, C++, SQL, JavaScript, Kotlin
ML & DL	PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, Pandas, SciPy, OpenCV
Visualization	Matplotlib, Seaborn, ggplot2
Developer Tools	Git, GitHub, Conda, Bash, Overleaf/LaTeX, Markdown
HPC & Parallel	Slurm (sbatch), IBM LSF (bsub), Indiana Jetstream, Distributed Data Parallel (DDP), Mixed-Precision Training (AMP)
GPU Computing	NVIDIA A100, H100, GH200, RTX 5090, RTX 3080, A6000, Pro 6000; CUDA, Triton kernel development
Languages	English (Native), Korean (Native), German (Conversational)

AWARDS AND HONORS

1. Computer Science Senior-Year Prize *Institutional*
   Bowdoin College May 2026
   - Awarded to a senior judged by the Department of Computer Science to have achieved the highest distinction in the major program in computer science.
2. Hastings Student AI Grant ($ 330) *Institutional*
   Bowdoin College Hastings - Initiative for AI and Humanity (HIAIH) March 2026
   - Awarded to support advanced GPU computing training through NVIDIA's Deep Learning Institute, with a focus on CUDA/C++ acceleration techniques applied to the Convolutional Nearest Neighbors senior honors project.
3. CRA Outstanding Undergraduate Researcher Award, Honorable Mention *National*
   Computing Research Association January 2026
   - Prestigious national honor recognizing exceptional undergraduate computing researchers; first Bowdoin student to receive this award. Website Link
4. John L. Roberts Fall Research Award ($ 2,463) *Institutional*
   Bowdoin College October 2025
   - Awarded for senior honors project on Convolutional Nearest Neighbors & Convolutional Nearest Neighbors Attention.
5. Best Poster Presentation Award ($ 100) *Institutional*
   Washington University in St. Louis July 2025
   - Awarded for best poster presentation among McKelvey Engineering summer research fellows at Washington University in St. Louis.
6. McKelvey Engineering Summer Research Fellowship ($ 7,200) *Institutional*
   Washington University in St. Louis May 2025
   - Awarded to conduct 10-week full-time research with the Computational Imaging Group.
7. Allen B. Tucker Computer Science Research Prize ($ 50) *Institutional*
   Bowdoin College May 2025
   - Awarded to a computer science student for excellence in summer research.
8. CRA Undergraduate Research Award ($ 4,000) *Local*
   Last Mile Education April 2025
   - Awarded to support summer research costs for computational imaging research at Washington University in St. Louis.
9. NYC STEM Award ($ 1,500) *Local*
   Last Mile Education March 2025
   - Merit-based grant for technical equipment and research support.
10. Google AI 2024 Award ($ 595) *Local*
    Last Mile Education January 2025
    - Awarded for independent semester research extending prior work from the Christenfeld Summer Research Fellowship.
11. Christenfeld Summer Research Fellowship ($ 4,800) *Institutional*
    Bowdoin College April 2024
    - Awarded to conduct 8-week full-time research in computer vision and deep learning under faculty mentorship.
12. NSF Student Faculty Research Fellowship ($ 1,800) *National/Institutional*
    Bowdoin College, National Science Foundation January 2024
    - Awarded for computational cognitive science research focused on modeling search and retrieval within the mental lexicon.
13. QuestBridge National College Match Finalist *National*
    QuestBridge September 2021
    - Recognized as a finalist for the QuestBridge program, which connects high-achieving students from low-income, first-generation backgrounds with full scholarships to top colleges.